BorrowBox — Database Schema & Architecture Guide

Full-Stack Equipment Rental Marketplace | Complete Database Reference Manual

The **BorrowBox** relational database (borrowbox_db) is structured around 13 core tables designed to manage users, equipment listings, date availability, booking lifecycle state transitions, condition inspection reports, verified reviews, and disputes.

1. Database Engine & Migration Architecture

- **Database Engine:** MySQL 8.x (Primary Production) / H2 Embedded (In-Memory Development Fallback)
- **Flyway Migrations:** db/migration/V1__schema.sql, db/migration/V2__seed_data.sql
- **DDL & DML Scripts:** database/schema.sql, database/seed-data.sql
- **Web Administration Console:** http://localhost:8080/h2-console (JDBC URL: jdbc:h2:mem:borrowbox_db)

2. Relational Database Tables Breakdown (13 Core Tables)

Table Name	Description & Function	Key Attributes & Constraints
users	Stores user profiles, credentials & roles	id, full_name, email (UNIQUE), phone, password_hash, role (ROLE_RENTER, ROLE_OWNER, ROLE_ADMIN)
categories	Equipment categories & icon definitions	id, name, slug (UNIQUE), description, icon_name, active
equipment	Gear listings, pricing & rental terms	id, owner_id (FK), category_id (FK), name, slug, daily_price, security_deposit, item_condition, status
equipment_images	Product gallery photos & ordering	id, equipment_id (FK), file_path, display_order, is_primary
bookings	Rental reservations & state machine	id, booking_reference (UNIQUE), equipment_id (FK), renter_id (FK), owner_id (FK), start_date, end_date, total_amount, booking_status
booking_status_history	Audit log for booking state changes	id, booking_id (FK), previous_status, new_status, changed_by_id (FK), notes
condition_reports	Pre-handover & post-return reports	id, booking_id (FK), reporter_id (FK), inspection_stage (BEFORE_HANDOVER, AFTER_RETURN), condition_notes, damage_notes
condition_report_images	Inspection photo evidence	id, condition_report_id (FK), file_path
wishlists	Saved wishlist items per user	id, user_id (FK), equipment_id (FK), UK(user_id, equipment_id)
reviews	Verified rental ratings & comments	id, booking_id (FK, UNIQUE), equipment_id (FK), renter_id (FK), rating (1-5), comment
notifications	In-app notification system	id, user_id (FK), title, message, type, is_read
disputes	Moderation tickets for lost/damaged gear	id, booking_id (FK), raised_by_id (FK), reason, description, status (OPEN, RESOLVED)
audit_logs	System operational activity logs	id, user_id, action, entity_type, entity_id, ip_address

3. Booking Lifecycle State Machine

Bookings strictly transition through defined states to prevent double-booking and ensure condition verification:
PENDING → **APPROVED** / **REJECTED** → **READY_FOR_PICKUP** → **ACTIVE** → **RETURN_REQUESTED** → **COMPLETED** / **CANCELLED** / **DISPUTED**

4. Useful SQL Administration Queries

- **Check Conflicting Booking Dates for Equipment ID 1:**

```
SELECT * FROM bookings
WHERE equipment_id = 1
AND booking_status IN ('APPROVED', 'ACTIVE')
AND (start_date <= '2026-10-20' AND end_date >= '2026-10-15');
```

- **Fetch Equipment Summary with Category and Owner Names:**

```
SELECT e.id, e.name, c.name AS category, u.full_name AS owner, e.daily_price, e.status
FROM equipment e
JOIN categories c ON e.category_id = c.id
JOIN users u ON e.owner_id = u.id;
```

- **Fetch Side-by-Side Pre & Post Inspection Reports for Booking #2:**

```
SELECT inspection_stage, condition_notes, damage_notes, confirmed_at
FROM condition_reports
WHERE booking_id = 2
ORDER BY id ASC;
```